

What Makes a Good Wine?

By: Jared Martin

2024-11-27

Background

The wine industry is a blend of agriculture, science, art and business. The industry is made up of thousands of vineyards across the globe with each wine-producing country or region having its own unique climates and soil types. These factors, along with the type of grape grown and how the wine was aged, can contribute significantly to the taste and quality of the wine. The sheer variety and complexity of wine makes confidently selecting a high-quality wine for a weekend meal or a special celebration a challenge for consumers.

To aid consumers in choosing a high-quality wine, this paper explores using machine learning methods to create a model to predict a wine's quality based on its chemical make-up and examining the characteristics of a high-quality wine.

Data Set

To model wine quality by chemical characteristics, this paper utilized data sets on Wine Quality donated to the UC Irvine Machine Learning Repository¹. The donated data included two data sets which consisted of red and white wines sampled from northern Portugal from 2004 to 2007 by the local wine certification agency CVRVV. Across the two data sets, there were 6497 observations of wines sampled. For privacy reasons, the data did not include grape type, wine brand, selling price, or vineyard name.

Both data sets did include 11 quantitative variables measuring certain chemical aspects for each wine sampled, such as the amount of sulfates, alcohol level, or pH of each wine. The data set also included one categorical ordinal variable measuring the quality of the wine. To measure the quality of wine, at least 3 taste testers had to evaluate each wine on a scale of 1 to 10. The final quality score is the median of all the taste testers' quality scores. Given that the quality measurement is the median score, the lowest level of quality is 3 and the highest is 9. Quality will be the dependent variable in the prediction model.

Exploratory Data Analysis

A table with the summary statistics and the results for the Shapiro-Wiles test for normality for all of the variables can be found in the following table. Based on the Shapiro-Wiles test calculated for all of the variables, the categorical variable "quality" follows a fairly normal distribution; therefore, for the prepossess of the following clustering analysis, "quality" was treated as a numeric value.

Identifying Key Characteristics of High Quality Wine Using Clustering

Wines of very high and low quality have very few observations. To better identify characteristics of high-quality wines, K-Means clustering was used to group the sample wines into 5 different clusters of roughly

¹Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Wine Quality [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C56S3T>.

equal size. Because quality follows a normal distribution, the clusters with the highest average quality score should have the highest portion of high quality wine.

Clustering revealed that wines with higher quality tend to have on average higher level of alcohol, and lower levels of sugar, chlorides, acidity and sulfates than wines of lesser quality wines.

To determine if the difference in cluster means was statistical significantly between the lowest and the highest quality clusters, a simple test was run across all of the variables. Below is a summary table of the results by independent variables. Across all 11 independent variables, the cluster mean for both groups were significant at a alpha level of .01.

Machine Learning Methods Used to Predict Quality

Given that there is more than two level in our dependent variable, quality this paper explored the use of K-Nearest Neighbors, Classification Trees, Random Forest, Support Vector Machines and Artificial Neural Network to predict the quality of the wine.

Cross Validation Methods Used to Evaluate the models

To evaluate each method, the Validation Set method was used to measure the testing classification and error rates. In addition, confusions tables were used to analysis how each machine learning method performed at classifying wines into the 7 quality categories.

Since the quality variable is ordinal the Mean Absolute Deviation (MAD) was also used to evaluate the accuracy of the machine learning methods. MAD preserve the order of the quality variable and measures how far off the predictions were when they miss the target. The lower MAD value a model has the better. MAD was calculated using the following formula.